Syllabus: Connectivity, Matching, Coloring.

Mark Distribution in Previous GATE

Year	2026 - 1	2026 - 2	2025 - 1	2025 - 2	2024 - 1	2024 - 2	2023	2022	2021 - 1	2021 - 2	Minimum
1 Mark Count	0	0	0	0	1	1	0	1	1	0	0
2 Marks Count	3	1	0	0	1	2	1	3	1	0	0
Total Marks	6	2	0	0	3	5	2	7	3	0	**0**

Welcome to the Graph Theory chapter, a cornerstone of Discrete Mathematics and a frequently tested area in the GATE Computer Science examination. Graph Theory provides a powerful framework for modeling relationships and processes across various domains, from social networks and transportation systems to computer networks and algorithm design. For GATE CS, this subject typically carries a weightage of **5-8 marks**, often integrated with Data Structures and Algorithms. Questions range from direct application of definitions and formulas, analysis of graph properties, tracing graph algorithms, to problem-solving scenarios involving connectivity, coloring, and planarity. A strong grasp of graph theory fundamentals is crucial not only for scoring well but also for understanding advanced topics in computer science.

Topic-wise Key Concepts

Counting

Counting in graph theory involves determining the number of possible graphs, paths, cycles, or specific structures within graphs. It often relies on fundamental combinatorial principles.

- **Definition and Core Idea:** This topic focuses on enumerating different types of graphs or substructures within them, using principles of permutations, combinations, and other combinatorial techniques. It's about understanding how many ways a graph can be formed or how many specific patterns exist.
- **Important Formulas, Theorems, and Results:**
  1. **Number of simple labeled graphs with n vertices:**

     $$2^{\binom{n}{2}} = 2^{n(n-1)/2}$$

     This is because for each pair of n vertices, there can either be an edge or not.
  2. **Number of simple directed labeled graphs with n vertices:**

     $$2^{n(n-1)}$$

     For each ordered pair of distinct vertices (u, v), there can be an edge from u to v, or not.
  3. **Number of spanning trees in a complete graph K_n (Cayley's Formula):**

     $$n^{n-2}$$

     This formula is critical for counting trees on a given set of labeled vertices.
  4. **Number of spanning trees in a general graph (Matrix Tree Theorem):** This theorem uses the Laplacian matrix of a graph, but its direct application is usually beyond GATE scope; Cayley's formula is more common.
  5. **Number of paths of length k between two vertices in an adjacency matrix:** The (i, j)-th entry of A^k (where A is the adjacency matrix) gives the number of paths of length k from vertex i to vertex j.
- **Key Properties and Identities:**
  - Understanding the difference between labeled and unlabeled graphs is crucial. GATE questions usually imply labeled graphs unless specified.
  - Basic combinatorial identities like $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ and $P(n, k) = \frac{n!}{(n-k)!}$ are fundamental.
  - The Inclusion-Exclusion Principle can be used for more complex counting problems, especially when dealing with properties that overlap.
- **Common Pitfalls or Tricky Points:**
  - **Overcounting or Undercounting:** Ensure each distinct configuration is counted exactly once.
  - **Distinguishing Labeled vs. Unlabeled:** Most GATE questions assume labeled vertices unless stated otherwise, which simplifies counting. Unlabeled graph counting is significantly harder.
  - **Specific Graph Types:** Be careful when counting paths/cycles in specific graph types like trees, complete graphs, or bipartite graphs.

- **Standard Problem-Solving Techniques or Shortcuts:**
  - **Direct Application of Formulas:** For simple graphs, complete graphs, and spanning trees, direct formula application is common.
  - **Combinatorial Arguments:** Break down the problem into smaller, manageable choices and multiply/add the possibilities.
  - **Recurrence Relations:** For counting paths or specific structures, sometimes a recurrence relation can be formulated and solved.

Degree of Graph

The degree of a vertex is a fundamental property that quantifies its connectivity within a graph. It's essential for understanding graph structure and validating basic graph properties.

- **Definition and Core Idea:** The degree of a vertex v, denoted $\deg(v)$ or $d(v)$, is the number of edges incident to it. For directed graphs, we distinguish between in-degree (edges pointing to v) and out-degree (edges pointing from v).
- **Important Formulas, Theorems, and Results:**
  1. **Handshaking Lemma (for undirected graphs):** The sum of the degrees of all vertices in any undirected graph is equal to twice the number of edges.

     $$\sum_{v \in V} \deg(v) = 2|E|$$

     This is a cornerstone theorem in graph theory.
  2. **For directed graphs:** The sum of in-degrees equals the sum of out-degrees, and both are equal to the total number of edges.

     $$\sum_{v \in V} \text{in-deg}(v) = \sum_{v \in V} \text{out-deg}(v) = |E|$$

  3. **Number of odd-degree vertices:** In any undirected graph, the number of vertices with odd degree must be even. This is a direct consequence of the Handshaking Lemma.
  4. **Maximum degree $\Delta(G)$ and Minimum degree $\delta(G)$:** These denote the maximum and minimum degrees among all vertices in graph G, respectively.
- **Key Properties and Identities:**
  - The Handshaking Lemma is a powerful tool for checking the validity of a degree sequence or finding the number of edges.
  - A graph cannot exist if its degree sequence violates the Handshaking Lemma or has an odd number of odd-degree vertices.
  - For a simple graph with n vertices, the maximum degree of any vertex is $n - 1$.
- **Common Pitfalls or Tricky Points:**
  - **Self-loops:** In some contexts, a self-loop at a vertex contributes 2 to its degree. GATE questions usually specify if self-loops are allowed or assume simple graphs (no self-loops, no multiple edges).
  - **Multiple Edges:** If multiple edges are allowed between two vertices, each edge contributes to the degree.
  - **Directed vs. Undirected:** Always be clear whether the graph is directed or undirected when calculating degrees.
- **Standard Problem-Solving Techniques or Shortcuts:**
  - **Direct Application of Handshaking Lemma:** Often used to find the number of edges given a degree sequence, or to determine if a given degree sequence is possible.
  - **Parity Check:** Quickly identify impossible degree sequences by checking the number of odd-degree vertices.

Graph Algorithms

Graph algorithms are systematic procedures for solving problems on graphs, such as finding paths, spanning trees, or optimal routes. They are central to computer science applications.

- **Definition and Core Idea:** These are algorithms designed to traverse, search, analyze, and optimize properties of graphs. They form the backbone of many real-world applications, from network routing to social network analysis.
- **Important Algorithms, Formulas, and Complexities:**
  1. **Breadth-First Search (BFS):** Explores a graph layer by layer.
     - **Purpose:** Find shortest path in unweighted graphs, find connected components.
     - **Time Complexity:** $O(V + E)$ for adjacency list, $O(V^2)$ for adjacency matrix.
  2. **Depth-First Search (DFS):** Explores as far as possible along each branch before backtracking.
     - **Purpose:** Find connected components, topological sort, detect cycles, find bridges/articulation points.

        - **Time Complexity:** $O(V+E)$ for adjacency list, $O(V^2)$ for adjacency matrix.
    3. **Dijkstra's Algorithm:** Finds the shortest paths from a single source vertex to all other vertices in a graph with non-negative edge weights.
        - **Purpose:** Single-source shortest path.
        - **Time Complexity:** $O(E \log V)$ with a Fibonacci heap, $O(E + V \log V)$ with a binary heap, $O(V^2)$ with adjacency matrix.
    4. **Prim's Algorithm:** Finds a Minimum Spanning Tree (MST) in a weighted, undirected graph.
        - **Purpose:** MST.
        - **Time Complexity:** $O(E \log V)$ with a binary heap, $O(V^2)$ with adjacency matrix.
    5. **Kruskal's Algorithm:** Finds a Minimum Spanning Tree (MST) in a weighted, undirected graph.
        - **Purpose:** MST.
        - **Time Complexity:** $O(E \log E)$ or $O(E \log V)$ using a Disjoint Set Union (DSU) data structure.
    6. **Bellman-Ford Algorithm:** Finds the shortest paths from a single source vertex to all other vertices, even with negative edge weights. Detects negative cycles.
        - **Purpose:** Single-source shortest path with negative weights.
        - **Time Complexity:** $O(VE)$.
    7. **Floyd-Warshall Algorithm:** Finds all-pairs shortest paths in a weighted graph, can handle negative weights but not negative cycles.
        - **Purpose:** All-pairs shortest path.
        - **Time Complexity:** $O(V^3)$.
    8. **Topological Sort:** Linear ordering of vertices such that for every directed edge (u, v), u comes before v. Applicable only to Directed Acyclic Graphs (DAGs).
        - **Purpose:** Ordering tasks with dependencies.
        - **Time Complexity:** $O(V+E)$ (using DFS or Kahn's algorithm).
    9. **Ford-Fulkerson Algorithm (and Edmonds-Karp):** Finds the maximum flow in a flow network.
        - **Purpose:** Maximum flow.
        - **Time Complexity:** Ford-Fulkerson is not polynomial in general; Edmonds-Karp is $O(VE^2)$.
- **Key Properties and Identities:**
    - **Greedy Algorithms:** Prim's, Kruskal's, Dijkstra's are greedy algorithms that make locally optimal choices.
    - **Dynamic Programming:** Floyd-Warshall and Bellman-Ford use dynamic programming principles.
    - **Connectivity:** BFS/DFS can determine if a graph is connected or find its connected components.
    - **Cycles:** DFS can detect cycles in both directed and undirected graphs.
- **Common Pitfalls or Tricky Points:**
    - **Negative Edge Weights:** Dijkstra's fails with negative edge weights; use Bellman-Ford or Floyd-Warshall.
    - **Disconnected Graphs:** Algorithms like Dijkstra's might not reach all vertices if the graph is disconnected from the source.
    - **Choosing the Right Algorithm:** Understand the constraints (weighted/unweighted, directed/undirected, negative weights, single-source/all-pairs) to select the appropriate algorithm.
    - **Adjacency List vs. Matrix:** Be aware of how the graph representation affects time complexity.
- **Standard Problem-Solving Techniques or Shortcuts:**
    - **Tracing:** Practice tracing algorithms step-by-step on small graphs to understand their mechanics.
    - **Recognizing Patterns:** Identify if a problem can be mapped to a standard graph algorithm (e.g., "minimum cost to connect all cities" suggests MST).
    - **Complexity Analysis:** Be able to derive or recall the time and space complexity for common algorithms.

Graph Coloring

Graph coloring assigns labels (colors) to graph elements subject to certain constraints, often used to model scheduling or resource allocation problems.

- **Definition and Core Idea:** Vertex coloring assigns a color to each vertex such that no two adjacent vertices have the same color. The minimum number of colors required for a graph G is called its chromatic number, denoted $\chi(G)$.
- **Important Formulas, Theorems, and Results:**
    1. **Chromatic Number $\chi(G)$:** The smallest integer k such that G is k-colorable.
    2. **Lower Bound:** For any graph G, $\chi(G) \geq \omega(G)$, where $\omega(G)$ is the clique number (size of the largest clique).
    3. **Upper Bound (Brooks' Theorem):** For a connected graph G that is neither a complete graph nor an odd cycle, $\chi(G) \leq \Delta(G)$, where $\Delta(G)$ is the maximum degree. In general, $\chi(G) \leq \Delta(G) + 1$.
    4. **Bipartite Graphs:** A graph is bipartite if and only if it contains no odd-length cycles. If a graph is bipartite and has at least one edge, then $\chi(G) = 2$.

5. **Planar Graphs (Four Color Theorem):** Every planar graph is 4-colorable. Thus, for any planar graph G, $\chi(G) \leq 4$.
6. **Chromatic Polynomial $P(G, k)$:** A polynomial whose value at integer k is the number of ways to color G using at most k colors. For a graph with n vertices, $P(G, k)$ is a polynomial in k of degree n.

- **Key Properties and Identities:**
  - A graph is 2-colorable if and only if it is bipartite.
  - If a graph contains a clique of size k, then at least k colors are needed.
  - The chromatic number of a complete graph K_n is n.
  - The chromatic number of a cycle graph C_n is 2 if n is even, and 3 if n is odd.
- **Common Pitfalls or Tricky Points:**
  - **Minimum vs. Any Coloring:** The goal is to find the minimum number of colors, not just any valid coloring.
  - **Identifying Cliques:** Finding the largest clique ($\omega(G)$) can be hard, but it provides a lower bound for $\chi(G)$.
  - **Edge Coloring ($\chi'(G)$):** Similar concept but for edges (adjacent edges have different colors). Vizing's Theorem states $\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1$. GATE usually focuses on vertex coloring.
- **Standard Problem-Solving Techniques or Shortcuts:**
  - **Greedy Coloring:** Assign colors iteratively to vertices in some order (e.g., decreasing degree). While not always optimal, it's a common heuristic.
  - **Identify Cliques and Odd Cycles:** These structures directly give lower bounds for the chromatic number.
  - **Trial and Error/Backtracking:** For small graphs, try to color with 2, then 3, etc., until a valid coloring is found.

Graph Connectivity

Graph connectivity measures how robust a graph is to the removal of vertices or edges. It's crucial for network reliability and fault tolerance.

- **Definition and Core Idea:** Connectivity refers to the property of a graph being "connected," meaning there's a path between any two vertices. More generally, it quantifies how many vertices or edges must be removed to disconnect the graph.
- **Important Formulas, Theorems, and Results:**
  1. **Connected Graph:** A graph is connected if there is a path between every pair of distinct vertices.
  2. **Connected Components:** Maximal connected subgraphs.
  3. **Cut Vertex (Articulation Point):** A vertex whose removal increases the number of connected components.
  4. **Cut Edge (Bridge):** An edge whose removal increases the number of connected components.
  5. **Vertex Connectivity $\kappa(G)$:** The minimum number of vertices whose removal disconnects G or reduces it to a trivial graph. A graph is k-connected if $\kappa(G) \geq k$.
  6. **Edge Connectivity $\lambda(G)$:** The minimum number of edges whose removal disconnects G. A graph is k-edge-connected if $\lambda(G) \geq k$.
  7. **Relationship:** For any graph G, $\kappa(G) \leq \lambda(G) \leq \delta(G)$, where $\delta(G)$ is the minimum degree of G.
  8. **Menger's Theorem:** For any two non-adjacent vertices u and v in a graph G, the maximum number of vertex-disjoint paths between u and v is equal to the minimum number of vertices whose removal separates u and v. (Similar theorem for edge-disjoint paths and edge cuts).
- **Key Properties and Identities:**
  - A tree with more than one vertex has $\kappa(G) = 1$ and $\lambda(G) = 1$.
  - A complete graph K_n has $\kappa(K_n) = n - 1$ and $\lambda(K_n) = n - 1$.
  - If a graph has a bridge, its edge connectivity is 1. If it has an articulation point, its vertex connectivity is 1 (unless it's K_2).
- **Common Pitfalls or Tricky Points:**
  - **Trivial Graph:** For a complete graph K_n, removing $n - 1$ vertices disconnects it into a trivial graph (single vertex), so $\kappa(K_n) = n - 1$.
  - **Distinguishing Vertex vs. Edge Connectivity:** Carefully read whether the question asks for vertex or edge removal.
  - **Disconnected Graphs:** For a disconnected graph, $\kappa(G) = 0$ and $\lambda(G) = 0$.
- **Standard Problem-Solving Techniques or Shortcuts:**
  - **BFS/DFS:** Can be used to find connected components, bridges, and articulation points.
  - **Trial and Error:** For small graphs, try removing vertices/edges to find the minimum set that disconnects the graph.
  - **Apply $\kappa(G) \leq \lambda(G) \leq \delta(G)$:** This inequality provides quick bounds.

Graph Isomorphism

Graph isomorphism determines if two graphs are structurally identical, regardless of their vertex labeling or drawing style. It's a fundamental concept for comparing graphs.

- **Definition and Core Idea:** Two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic if there exists a bijective function $f : V_1 \to V_2$ such that for any pair of vertices $u, v \in V_1$, $\{u, v\} \in E_1$ if and only if $\{f(u), f(v)\} \in E_2$. Essentially, they have the same structure.
- **Important Properties (Invariants):**
  1. **Number of Vertices:** $|V_1| = |V_2|$.
  2. **Number of Edges:** $|E_1| = |E_2|$.
  3. **Degree Sequence:** Both graphs must have the same degree sequence (multiset of degrees).
  4. **Number of Connected Components:** Must be the same.
  5. **Cycle Lengths:** The number of cycles of each specific length must be the same.
  6. **Connectivity:** Same vertex and edge connectivity.
  7. **Chromatic Number:** Must be the same.
  8. **Presence of Subgraphs:** If G_1 contains a subgraph H, then G_2 must also contain H.
- **Key Properties and Identities:**
  - If any of the invariants differ, the graphs are definitely NOT isomorphic.
  - If all invariants are the same, the graphs MIGHT be isomorphic, but it's not a guarantee (e.g., regular graphs can have the same degree sequence but be non-isomorphic).
  - The Graph Isomorphism Problem (determining if two graphs are isomorphic) is in NP, but its exact complexity class is unknown (neither P nor NP-complete is proven). For GATE, problems are usually small enough to solve by inspection or invariant checking.
- **Common Pitfalls or Tricky Points:**
  - **Proving Non-Isomorphism:** This is generally easier; just find one invariant that differs.
  - **Proving Isomorphism:** This requires finding an explicit bijective mapping f, which can be challenging. Often, it involves matching vertices with identical properties (e.g., same degree, same neighborhood structure).
  - **Visual Deception:** Graphs drawn differently can be isomorphic, and graphs drawn similarly can be non-isomorphic.
- **Standard Problem-Solving Techniques or Shortcuts:**
  - **Check Invariants Systematically:** Start with simple invariants like number of vertices, edges, and degree sequence. If any differ, stop.
  - **Match Vertices by Properties:** Look for unique vertices (e.g., a vertex with degree 1, or a vertex part of a unique cycle structure) and try to map them.
  - **Adjacency Matrix Comparison:** If two graphs are isomorphic, their adjacency matrices can be made identical by permuting rows and columns. This is computationally intensive for humans but conceptually useful.

Graph Matching

Graph matching is about finding a set of edges that do not share any common vertices, often used in assignment problems or resource allocation.

- **Definition and Core Idea:** A matching M in a graph G is a set of edges such that no two edges in M share a common vertex. A vertex is *matched* (or covered) if it is an endpoint of an edge in M; otherwise, it is *unmatched*.
- **Important Formulas, Theorems, and Results:**
  1. **Maximum Matching:** A matching with the largest possible number of edges.
  2. **Perfect Matching:** A matching that covers all vertices of the graph. A perfect matching can only exist if $|V|$ is even.
  3. **Augmenting Path:** A path that starts and ends at unmatched vertices, and whose edges alternate between being in the current matching and not in the current matching. An augmenting path can be used to increase the size of a matching.
  4. **Hall's Marriage Theorem (for Bipartite Graphs):** A bipartite graph $G = (U \cup V, E)$ has a matching that covers all vertices in U if and only if for every subset $A \subseteq U$, $|N(A)| \geq |A|$, where $N(A)$ is the set of neighbors of vertices in A.
  5. **Tutte's Theorem (for General Graphs):** A graph G has a perfect matching if and only if for every subset $S \subseteq V$, the number of odd components in $G - S$ is less than or equal to $|S|$. (This is more complex and less common in GATE than Hall's).
- **Key Properties and Identities:**
  - The size of a maximum matching in a bipartite graph can be found using network flow algorithms (e.g., by converting to a max-flow problem).
  - Every perfect matching is a maximum matching, but not vice-versa.
  - A graph can have multiple maximum matchings.

- **Common Pitfalls or Tricky Points:**
  - **Confusing Matching with Vertex Cover or Edge Cover:** These are related but distinct concepts. A vertex cover is a set of vertices that touch all edges. An edge cover is a set of edges that touch all vertices (only exists if no isolated vertices).
  - **General vs. Bipartite Graphs:** Finding maximum matchings in general graphs is more complex (e.g., Edmonds' blossom algorithm) than in bipartite graphs. GATE usually focuses on bipartite matching or simpler cases.
  - **Existence of Perfect Matching:** A perfect matching requires an even number of vertices.
- **Standard Problem-Solving Techniques or Shortcuts:**
  - **Greedy Approach:** While not always optimal for maximum matching, a greedy approach (e.g., pick an edge, remove its endpoints, repeat) can give a good starting point.
  - **Augmenting Paths:** For bipartite graphs, the concept of augmenting paths is key. Start with an empty matching, find an augmenting path, update the matching, and repeat until no more augmenting paths exist. This is the basis of the Hopcroft-Karp algorithm.
  - **Applying Hall's Theorem:** For bipartite graphs, use Hall's condition to check for the existence of a matching covering one side.

Graph Planarity

Graph planarity deals with whether a graph can be drawn on a plane without any edges crossing. It has applications in circuit design and network visualization.

- **Definition and Core Idea:** A graph is planar if it can be drawn in the plane without any edges crossing each other (except at vertices). Such a drawing is called a planar embedding.
- **Important Formulas, Theorems, and Results:**
  1. **Euler's Formula for Planar Graphs:** For any connected planar graph with V vertices, E edges, and F faces (regions),
  $$V - E + F = 2$$
  This formula is fundamental for planar graphs.
  2. **Corollaries of Euler's Formula (for simple connected planar graphs with $V \geq 3$):**
     - If G has no cycles of length 3 (i.e., no K_3 subgraph), then $E \leq 2V - 4$.
     - In general, $E \leq 3V - 6$.
     - There is at least one vertex with degree at most 5 ($\delta(G) \leq 5$).
  3. **Kuratowski's Theorem:** A graph is planar if and only if it does not contain a subgraph that is a subdivision of K_5 (complete graph on 5 vertices) or $K_{3,3}$ (complete bipartite graph with 3 vertices in each partition).
     - A subdivision of a graph is obtained by replacing edges with paths.
- **Key Properties and Identities:**
  - K_5 and $K_{3,3}$ are non-planar.
  - Any graph that contains K_5 or $K_{3,3}$ as a minor (or subdivision) is non-planar.
  - The dual of a planar graph is also planar.
- **Common Pitfalls or Tricky Points:**
  - **Misapplying Euler's Formula:** Remember it's for *connected* planar graphs. For disconnected graphs, $V - E + F = 1 + k$, where k is the number of connected components.
  - **Identifying Subdivisions:** Recognizing K_5 or $K_{3,3}$ subdivisions can be tricky. Look for 5 vertices all connected to each other (possibly via paths) or 6 vertices arranged in a bipartite manner.
  - **Different Drawings:** A graph might look non-planar in one drawing but be planar in another. The definition requires *existence* of a planar drawing.
- **Standard Problem-Solving Techniques or Shortcuts:**
  - **Check Euler's Formula Corollaries:** If $E > 3V - 6$ (or $E > 2V - 4$ if no triangles), the graph is definitely non-planar. This is a quick check for non-planarity.
  - **Look for K_5 or $K_{3,3}$ Subdivisions:** Try to simplify the graph by removing degree-2 vertices (which don't affect planarity for Kuratowski's theorem) or identifying dense subgraphs.
  - **Redrawing:** For small graphs, try to redraw them to eliminate crossings.

Jaccard Coefficient

The Jaccard Coefficient is a statistical measure used to gauge the similarity between two sets, finding applications in data mining, machine learning, and graph analysis (e.g., comparing neighbor sets).

- **Definition and Core Idea:** The Jaccard Coefficient (or Jaccard Index) measures the similarity between two finite sets, A and B, as the size of their intersection divided by the size of their union. It quantifies the overlap between the sets.

- **Important Formulas, Theorems, and Results:**
  1. **Jaccard Coefficient $J(A,B)$:**

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|}$$

  2. **Jaccard Distance $D_J(A,B)$:** Measures dissimilarity.

$$D_J(A,B) = 1 - J(A,B) = \frac{|A \cup B| - |A \cap B|}{|A \cup B|}$$

- **Key Properties and Identities:**
  - The Jaccard Coefficient ranges from 0 to 1.
    - $J(A,B) = 1$ if $A = B$ (perfect similarity).
    - $J(A,B) = 0$ if $A \cap B = \emptyset$ (no similarity).
  - It is symmetric: $J(A,B) = J(B,A)$.
  - It is sensitive to the size of the union; a large union with a small intersection leads to a low coefficient.
- **Common Pitfalls or Tricky Points:**
  - **Miscalculating Intersection or Union:** Ensure correct set operations, especially when dealing with lists that might contain duplicates (convert to sets first).
  - **Empty Sets:** If both sets are empty, the Jaccard coefficient is usually defined as 1 (perfect similarity). If one is empty and the other is not, it's 0.
  - **Application Context:** In graph theory, it's often used to compare the neighborhood sets of two vertices to determine their "structural similarity" or to compare sets of features associated with graph nodes.
- **Standard Problem-Solving Techniques or Shortcuts:**
  - **Direct Calculation:** Identify the elements in the intersection and union, then apply the formula.
  - **Venn Diagrams:** For small sets, a Venn diagram can help visualize the intersection and union.

Quick Formula Reference

- **Counting Simple Labeled Graphs (n vertices):** $2^{\binom{n}{2}}$
- **Counting Simple Directed Labeled Graphs (n vertices):** $2^{n(n-1)}$
- **Cayley's Formula (Spanning Trees in K_n):** n^{n-2}
- **Handshaking Lemma (Undirected):** $\sum_{v \in V} \deg(v) = 2|E|$
- **Handshaking Lemma (Directed):** $\sum_{v \in V} \text{in-deg}(v) = \sum_{v \in V} \text{out-deg}(v) = |E|$
- **BFS/DFS Time Complexity:** $O(V+E)$ (adj list), $O(V^2)$ (adj matrix)
- **Dijkstra's Time Complexity:** $O(E \log V)$ (binary heap), $O(V^2)$ (adj matrix)
- **Prim's/Kruskal's Time Complexity (MST):** $O(E \log V)$ or $O(E \log E)$
- **Bellman-Ford Time Complexity:** $O(VE)$
- **Floyd-Warshall Time Complexity:** $O(V^3)$
- **Topological Sort Time Complexity:** $O(V+E)$
- **Chromatic Number Lower Bound:** $\chi(G) \geq \omega(G)$ (clique number)
- **Chromatic Number Upper Bound (Brooks' Theorem):** $\chi(G) \leq \Delta(G)$ (for non-complete, non-odd cycle connected graphs), else $\chi(G) \leq \Delta(G) + 1$
- **Bipartite Graph Chromatic Number:** $\chi(G) = 2$ (if connected and has edges)
- **Planar Graph Chromatic Number (Four Color Theorem):** $\chi(G) \leq 4$
- **Connectivity Relationship:** $\kappa(G) \leq \lambda(G) \leq \delta(G)$
- **Euler's Formula (Connected Planar Graph):** $V - E + F = 2$
- **Planarity Condition (Simple Connected Planar, $V \geq 3$):** $E \leq 3V - 6$
- **Planarity Condition (Simple Connected Planar, no K_3, $V \geq 3$):** $E \leq 2V - 4$
- **Kuratowski's Theorem:** Planar iff no K_5 or $K_{3,3}$ subdivision
- **Jaccard Coefficient:** $J(A,B) = \frac{|A \cap B|}{|A \cup B|}$
- **Jaccard Distance:** $D_J(A,B) = 1 - J(A,B)$

Important Tips for GATE

1. **Master Definitions and Basic Properties:** Many GATE questions test fundamental definitions (e.g., what is a bridge, what is a perfect matching) and direct consequences of theorems. Ensure you know what each term means precisely.
2. **Practice Drawing and Visualizing Graphs:** For problems involving planarity, isomorphism, or connectivity, sketching the graph and trying different layouts can be immensely helpful. Don't rely solely on abstract representations.
3. **Understand Algorithm Mechanics and Complexity:** Don't just memorize complexities; understand how BFS, DFS, Dijkstra's, Prim's, and Kruskal's work step-by-step. Be prepared to trace them on small examples and identify their time/space complexities for both adjacency list and matrix representations.
4. **Pay Attention to Graph Types and Constraints:** Is the graph directed or undirected? Weighted or unweighted? Simple or multi-graph? Does it allow self-loops? These details significantly impact which formulas or algorithms apply. For example, Dijkstra's fails with negative edge weights.
5. **Use Invariants for Isomorphism:** When checking for isomorphism, systematically compare invariants like number of vertices, edges, degree sequence, and cycle lengths. If any invariant differs, the graphs are not isomorphic. This is a powerful shortcut.
6. **Be Wary of Common Pitfalls:** Remember that Euler's formula applies to connected planar graphs. Dijkstra's doesn't work with negative weights. A greedy coloring isn't always optimal. Keep these exceptions in mind.
7. **Time Management for Tracing Problems:** For algorithm tracing questions, work carefully and systematically. Use scratch paper to keep track of visited nodes, distances, or parent pointers. Avoid rushing, as a single error can invalidate the entire trace.
8. **Review Special Graph Types:** Be familiar with properties of common graphs like complete graphs (K_n), complete bipartite graphs ($K_{m,n}$), cycle graphs (C_n), path graphs (P_n), and trees. Their specific properties often appear in questions.

2.1 Counting (3)

2.1.1 Counting: GATE CSE 2001 | Question: 2.15

How many undirected graphs (not necessarily connected) can be constructed out of a given set $V = \{v_1, v_2, \dots v_n\}$ of n vertices?

A. $\frac{n(n-1)}{2}$ B. 2^n C. $n!$ D. $2^{\frac{n(n-1)}{2}}$

gatecse-2001 graph-theory normal counting

Answer key☟

2.1.2 Counting: GATE CSE 2004 | Question: 79

How many graphs on n labeled vertices exist which have at least $\frac{(n^2-3n)}{2}$ edges ?

A. ${}^{\left(\frac{n^2-n}{2}\right)}C_{\left(\frac{n^2-3n}{2}\right)}$

B. $\sum_{k=0}^{\left(\frac{n^2-3n}{2}\right)} {}^{(n^2-n)}C_k$

C. ${}^{\left(\frac{n^2-n}{2}\right)}C_n$

D. $\sum_{k=0}^{n} {}^{\left(\frac{n^2-n}{2}\right)}C_k$

gatecse-2004 graph-theory combinatory normal counting

Answer key☟

2.1.3 Counting: GATE CSE 2012 | Question: 38

Let G be a complete undirected graph on 6 vertices. If vertices of G are labeled, then the number of distinct cycles of length 4 in G is equal to

A. 15 B. 30 C. 90 D. 360

gatecse-2012 graph-theory normal marks-to-all counting

Answer key☟

2.2 Degree of Graph (13)

2.2.1 Degree of Graph: GATE CSE 1987 | Question: 9c

Show that the number of odd-degree vertices in a finite graph is even.

gate1987 graph-theory degree-of-graph descriptive proof

Answer key

2.2.2 Degree of Graph: GATE CSE 1991 | Question: 16-b

Show that all vertices in an undirected finite graph cannot have distinct degrees, if the graph has at least two vertices.

gate1991 graph-theory degree-of-graph descriptive proof

Answer key

2.2.3 Degree of Graph: GATE CSE 1995 | Question: 24

Prove that in finite graph, the number of vertices of odd degree is always even.

gate1995 graph-theory degree-of-graph proof descriptive

Answer key

2.2.4 Degree of Graph: GATE CSE 2003 | Question: 40

A graph $G = (V, E)$ satisfies $|E| \leq 3 \mid V \mid -6$. The min-degree of G is defined as $\min_{v \in V} \{\text{degree } (v)\}$. Therefore, min-degree of G cannot be

A. 3

B. 4

C. 5

D. 6

gatecse-2003 graph-theory normal degree-of-graph

Answer key

2.2.5 Degree of Graph: GATE CSE 2006 | Question: 71

The 2^n vertices of a graph G corresponds to all subsets of a set of size n, for $n \geq 6$. Two vertices of G are adjacent if and only if the corresponding sets intersect in exactly two elements.
The number of vertices of degree zero in G is:

A. 1

B. n

C. $n+1$

D. 2^n

gatecse-2006 graph-theory normal degree-of-graph

Answer key

2.2.6 Degree of Graph: GATE CSE 2006 | Question: 72

The 2^n vertices of a graph G corresponds to all subsets of a set of size n, for $n \geq 6$. Two vertices of G are adjacent if and only if the corresponding sets intersect in exactly two elements.

The maximum degree of a vertex in G is:

A. $\binom{\frac{n}{2}}{2}.2^{\frac{n}{2}}$

B. 2^{n-2}

C. $2^{n-3} \times 3$

D. 2^{n-1}

gatecse-2006 graph-theory normal degree-of-graph

Answer key

2.2.7 Degree of Graph: GATE CSE 2009 | Question: 3

Which one of the following is **TRUE** for any simple connected undirected graph with more than 2 vertices?

A. No two vertices have the same degree.
B. At least two vertices have the same degree.
C. At least three vertices have the same degree.
D. All vertices have the same degree.

gatecse-2009 graph-theory normal degree-of-graph

Answer key

2.2.8 Degree of Graph: GATE CSE 2010 | Question: 1

Let $G = (V, E)$ be a graph. Define $\xi(G) = \sum_d i_d * d$, where i_d is the number of vertices of degree d in G. If S and T are two different trees with $\xi(S) = \xi(T)$, then

A. $|S| = 2|T|$
B. $|S| = |T| - 1$
C. $|S| = |T|$
D. $|S| = |T| + 1$

gatecse-2010 graph-theory normal degree-of-graph

Answer key

2.2.9 Degree of Graph: GATE CSE 2010 | Question: 28

The degree sequence of a simple graph is the sequence of the degrees of the nodes in the graph in decreasing order. Which of the following sequences can not be the degree sequence of any graph?

I. $7, 6, 5, 4, 4, 3, 2, 1$
II. $6, 6, 6, 6, 3, 3, 2, 2$
III. $7, 6, 6, 4, 4, 3, 2, 2$
IV. $8, 7, 7, 6, 4, 2, 1, 1$

A. I and II
B. III and IV
C. IV only
D. II and IV

gatecse-2010 graph-theory degree-of-graph

Answer key

2.2.10 Degree of Graph: GATE CSE 2013 | Question: 25

Which of the following statements is/are TRUE for undirected graphs?

P: Number of odd degree vertices is even.
Q: Sum of degrees of all vertices is even.

A. P only
B. Q only
C. Both P and Q
D. Neither P nor Q

gatecse-2013 graph-theory easy degree-of-graph

Answer key

2.2.11 Degree of Graph: GATE CSE 2014 | Set 1 | Question: 52

An ordered $n-$tuple $(d_1, d_2, \dots, d_n)$ with $d_1 \geq d_2 \geq \dots \geq d_n$ is called *graphic* if there exists a simple undirected graph with n vertices having degrees $d_1, d_2, \dots, d_n$ respectively. Which one of the following 6-tuples is NOT graphic?

A. $(1,1,1,1,1,1)$
B. $(2,2,2,2,2,2)$
C. $(3,3,3,1,0,0)$
D. $(3,2,1,1,1,0)$

gatecse-2014-set1 graph-theory normal degree-of-graph

Answer key

2.2.12 Degree of Graph: GATE CSE 2017 | Set 2 | Question: 23

G is an undirected graph with n vertices and 25 edges such that each vertex of G has degree at least 3. Then the maximum possible value of n is _________ .

gatecse-2017-set2 graph-theory numerical-answers degree-of-graph

Answer key

2.2.13 Degree of Graph: GATE CSE 2026 | Set 1 | Question: 37

Let $G(V, E)$ be a simple, undirected graph. A vertex cover of G is a subset $V' \subseteq V$ such that for every $(u, v) \in E, u \in V'$ or $v \in V'$. Let the size of the smallest vertex cover in G be k. Let S be any vertex cover of size k.

For a vertex $v \in V$, which of the following constraints will always ensure that $v \in S$?

A. The degree of v is at least $k+1$
B. The vertex v is on a path of length $k+1$
C. The vertex v is on a cycle of length $k+1$
D. The vertex v is a part of a clique of size k

gatecse-2026-set1 graph-theory two-marks degree-of-graph multiple-selects

Answer key☟

2.3 Graph Algorithms (1)

2.3.1 Graph Algorithms: GATE CSE 2026 | Set 1 | Question: 45

An undirected, unweighted, simple graph $G(V, E)$ is said to be 2-colorable if there exists a function $c : V \rightarrow \{0, 1\}$ such that for every $(u, v) \in E, c(u) \neq c(v)$.

Which of the following statements about 2-colorable graphs is/are true?

A. If G is 2-colorable, then G may contain cycles of odd length
B. If G is 2-colorable, then G may contain cycles of even length
C. An optimal algorithm for testing whether G is 2-colorable runs in time $\Theta(|V| + |E|)$, if G is represented as an adjacency list
D. An optimal algorithm for testing whether G is 2-colorable runs in time $\Theta(|E| \log |V|)$, if G is represented as an adjacency list

gatecse-2026-set1 graph-theory graph-algorithms two-marks multiple-selects

Answer key☟

2.4 Graph Coloring (11)

2.4.1 Graph Coloring: GATE CSE 2002 | Question: 1.4

The minimum number of colours required to colour the vertices of a cycle with n nodes in such a way that no two adjacent nodes have the same colour is

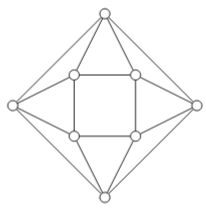
A. 2
B. 3
C. 4
D. $n - 2\left\lfloor \frac{n}{2} \right\rfloor + 2$

gatecse-2002 graph-theory graph-coloring normal

Answer key☟

2.4.2 Graph Coloring: GATE CSE 2004 | Question: 77

The minimum number of colours required to colour the following graph, such that no two adjacent vertices are assigned the same color, is



A. 2
B. 3
C. 4
D. 5

gatecse-2004 graph-theory graph-coloring easy

Answer key☟

2.4.3 Graph Coloring: GATE CSE 2009 | Question: 2

What is the chromatic number of an n vertex simple connected graph which does not contain any odd length cycle? Assume $n > 2$.

A. 2 B. 3 C. $n-1$ D. n

gatecse-2009 graph-theory graph-coloring normal

Answer key☟

2.4.4 Graph Coloring: GATE CSE 2016 | Set 2 | Question: 03

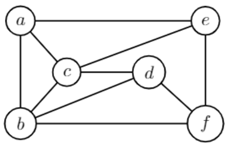
The minimum number of colours that is sufficient to vertex-colour any planar graph is ________.

gatecse-2016-set2 graph-theory graph-coloring normal numerical-answers

Answer key☟

2.4.5 Graph Coloring: GATE CSE 2018 | Question: 18

The chromatic number of the following graph is _____



graph-theory graph-coloring numerical-answers gatecse-2018 one-mark

Answer key☟

2.4.6 Graph Coloring: GATE CSE 2020 | Question: 52

Graph G is obtained by adding vertex s to $K_{3,4}$ and making s adjacent to every vertex of $K_{3,4}$. The minimum number of colours required to edge-colour G is _______

gatecse-2020 numerical-answers graph-theory graph-coloring two-marks

Answer key☟

2.4.7 Graph Coloring: GATE CSE 2023 | Question: 45

Let G be a simple, finite, undirected graph with vertex set $\{v_1, \ldots, v_n\}$. Let $\Delta(G)$ denote the maximum degree of G and let $\mathbb{N} = \{1, 2, \ldots\}$ denote the set of all possible colors. Color the vertices of G using the following greedy strategy: for $i = 1, \ldots, n$

$\text{color}(v_i) \leftarrow \min\{j \in \mathbb{N} : \text{no neighbour of } v_i \text{ is colored } j\}$

Which of the following statements is/are TRUE?

A. This procedure results in a proper vertex coloring of G.
B. The number of colors used is at most $\Delta(G) + 1$.
C. The number of colors used is at most $\Delta(G)$.
D. The number of colors used is equal to the chromatic number of G.

gatecse-2023 graph-theory graph-coloring multiple-selects two-marks

Answer key☟

2.4.8 Graph Coloring: GATE CSE 2024 | Set 1 | Question: 41

The chromatic number of a graph is the minimum number of colours used in a *proper* colouring of the graph. Let G be any graph with n vertices and chromatic number k. Which of the following statements is/are always TRUE?

A. G contains a complete subgraph with k vertices

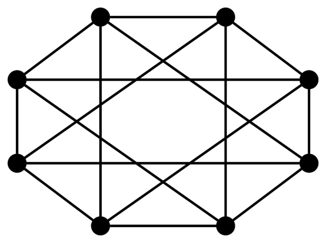
B. G contains an independent set of size at least n/k
C. G contains at least $k(k-1)/2$ edges
D. G contains a vertex of degree at least k

gatecse-2024-set1 multiple-selects graph-theory graph-coloring two-marks

Answer key

2.4.9 Graph Coloring: GATE CSE 2024 | Set 2 | Question: 50

The chromatic number of a graph is the minimum number of colours used in a proper colouring of the graph. The chromatic number of the following graph is __________.



gatecse-2024-set2 graph-theory numerical-answers graph-coloring two-marks

Answer key

2.4.10 Graph Coloring: GATE IT 2006 | Question: 25

Consider the undirected graph G defined as follows. The vertices of G are bit strings of length n. We have an edge between vertex u and vertex v if and only if u and v differ in exactly one bit position (in other words, v can be obtained from u by flipping a single bit). The ratio of the chromatic number of G to the diameter of G is,

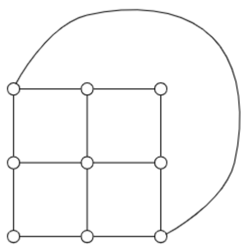
A. $\frac{1}{(2^{n-1})}$ B. $\left(\frac{1}{n}\right)$ C. $\left(\frac{2}{n}\right)$ D. $\left(\frac{3}{n}\right)$

gateit-2006 graph-theory graph-coloring normal

Answer key

2.4.11 Graph Coloring: GATE IT 2008 | Question: 3

What is the chromatic number of the following graph?



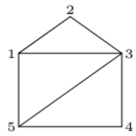
A. 2 B. 3 C. 4 D. 5

gateit-2008 graph-theory graph-coloring easy

Answer key

2.5 Graph Connectivity (40)

2.5.1 Graph Connectivity: GATE CSE 1987 | Question: 9d



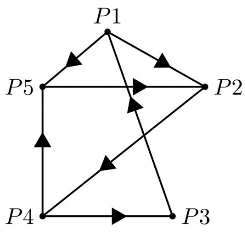
Specify an adjacency-lists representation of the undirected graph given above.

gate1987 graph-theory easy graph-connectivity descriptive

Answer key

2.5.2 Graph Connectivity: GATE CSE 1988 | Question: 2xvi

Write the adjacency matrix representation of the graph given in below figure.



gate1988 descriptive graph-theory graph-connectivity

Answer key

2.5.3 Graph Connectivity: GATE CSE 1990 | Question: 1-viii

A graph which has the same number of edges as its complement must have number of vertices congruent to ________ or ________ modulo 4.

gate1990 graph-theory graph-connectivity fill-in-the-blanks

Answer key

2.5.4 Graph Connectivity: GATE CSE 1991 | Question: 01,xv

The maximum number of possible edges in an undirected graph with n vertices and k components is ______.

gate1991 graph-theory graph-connectivity normal fill-in-the-blanks

Answer key

2.5.5 Graph Connectivity: GATE CSE 1992 | Question: 03,iii

How many edges can there be in a forest with p components having n vertices in all?

gate1992 graph-theory graph-connectivity descriptive

Answer key

2.5.6 Graph Connectivity: GATE CSE 1993 | Question: 8.1

Consider a simple connected graph G with n vertices and n edges $(n > 2)$. Then, which of the following statements are true?

A. G has no cycles
B. The graph obtained by removing any edge from G is not connected
C. G has at least one cycle
D. The graph obtained by removing any two edges from G is not connected
E. None of the above

gate1993 graph-theory graph-connectivity easy multiple-selects

Answer key

2.5.7 Graph Connectivity: GATE CSE 1994 | Question: 1.6, ISRO2008-29

The number of distinct simple graphs with up to three nodes is

A. 15 B. 10 C. 7 D. 9

gate1994 graph-theory graph-connectivity combinatory normal isro2008 counting

Answer key

2.5.8 Graph Connectivity: GATE CSE 1994 | Question: 2.5

The number of edges in a regular graph of degree d and n vertices is ____________

gate1994 graph-theory easy graph-connectivity fill-in-the-blanks

Answer key

2.5.9 Graph Connectivity: GATE CSE 1994 | Question: 24

An independent set in a graph is a subset of vertices such that no two vertices in the subset are connected by an edge. An incomplete scheme for a greedy algorithm to find a maximum independent set in a tree is given below:

```
V: Set of all vertices in the tree;
I := ϕ
while V ≠ ϕ do
begin
   select a vertex u ∊ V such that
   _______;
   V := V - {u};
   if u is such that
   ________then I := I ∪ {u}
end;
Output(I);
```

A. Complete the algorithm by specifying the property of vertex u in each case.
B. What is the time complexity of the algorithm?

gate1994 graph-theory normal descriptive graph-connectivity

Answer key

2.5.10 Graph Connectivity: GATE CSE 1995 | Question: 1.25

The minimum number of edges in a connected cyclic graph on n vertices is:

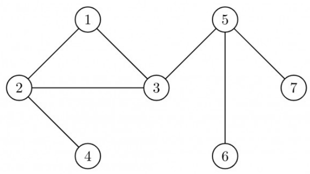
A. $n-1$ B. n C. $n+1$ D. None of the above

gate1995 graph-theory graph-connectivity easy

Answer key

2.5.11 Graph Connectivity: GATE CSE 1999 | Question: 1.15

The number of articulation points of the following graph is



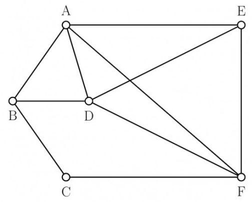
A. 0 B. 1 C. 2 D. 3

gate1999 graph-theory graph-connectivity normal

Answer key

2.5.12 Graph Connectivity: GATE CSE 1999 | Question: 5

Let G be a connected, undirected graph. A cut in G is a set of edges whose removal results in G being broken into two or more components, which are not connected with each other. The size of a cut is called its cardinality. A min-cut of G is a cut in G of minimum cardinality. Consider the following graph:



a. Which of the following sets of edges is a cut?
    i. $\{(A, B), (E, F), (B, D), (A, E), (A, D)\}$
    ii. $\{(B, D), (C, F), (A, B)\}$
b. What is cardinality of min-cut in this graph?
c. Prove that if a connected undirected graph G with n vertices has a min-cut of cardinality k, then G has at least $\left(\frac{n \times k}{2}\right)$ edges.

gate1999 graph-theory graph-connectivity normal descriptive proof

Answer key

2.5.13 Graph Connectivity: GATE CSE 2002 | Question: 1.25, ISRO2008-30, ISRO2016-6

The maximum number of edges in a n-node undirected graph without self loops is

A. n^2
B. $\frac{n(n-1)}{2}$
C. $n-1$
D. $\frac{(n+1)(n)}{2}$

gatecse-2002 graph-theory easy isro2008 isro2016 graph-connectivity

Answer key

2.5.14 Graph Connectivity: GATE CSE 2003 | Question: 8, ISRO2009-53

Let G be an arbitrary graph with n nodes and k components. If a vertex is removed from G, the number of components in the resultant graph must necessarily lie down between

A. k and n
B. $k-1$ and $k+1$
C. $k-1$ and $n-1$
D. $k+1$ and $n-k$

gatecse-2003 graph-theory graph-connectivity normal isro2009

Answer key

2.5.15 Graph Connectivity: GATE CSE 2004 | Question: 81

Let $G_1 = (V, E_1)$ and $G_2 = (V, E_2)$ be connected graphs on the same vertex set V with more than two vertices. If $G_1 \cap G_2 = (V, E_1 \cap E_2)$ is not a connected graph, then the graph $G_1 \cup G_2 = (V, E_1 \cup E_2)$

A. cannot have a cut vertex
B. must have a cycle
C. must have a cut-edge (bridge)
D. has chromatic number strictly greater than those of G_1 and G_2

gatecse-2004 graph-theory difficult graph-connectivity

Answer key

2.5.16 Graph Connectivity: GATE CSE 2005 | Question: 11

Let G be a simple graph with 20 vertices and 100 edges. The size of the minimum vertex cover of G is 8. Then, the size of the maximum independent set of G is:

A. 12
B. 8
C. less than 8
D. more than 12

gatecse-2005 graph-theory normal graph-connectivity

Answer key

2.5.17 Graph Connectivity: GATE CSE 2006 | Question: 73

The 2^n vertices of a graph G corresponds to all subsets of a set of size n, for $n \geq 6$. Two vertices of G are adjacent if and only if the corresponding sets intersect in exactly two elements.

The number of connected components in G is:

A. n
B. $n+2$
C. $2^{\frac{n}{2}}$
D. $\frac{2^n}{n}$

gatecse-2006 graph-theory normal graph-connectivity

Answer key

2.5.18 Graph Connectivity: GATE CSE 2007 | Question: 23

Which of the following graphs has an Eulerian circuit?

A. Any k-regular graph where k is an even number.
B. A complete graph on 90 vertices.
C. The complement of a cycle on 25 vertices.
D. None of the above

gatecse-2007 graph-theory normal graph-connectivity

Answer key

2.5.19 Graph Connectivity: GATE CSE 2008 | Question: 42

G is a graph on n vertices and $2n-2$ edges. The edges of G can be partitioned into two edge-disjoint spanning trees. Which of the following is NOT true for G?

A. For every subset of k vertices, the induced subgraph has at most $2k-2$ edges.
B. The minimum cut in G has at least 2 edges.
C. There are at least 2 edge-disjoint paths between every pair of vertices.
D. There are at least 2 vertex-disjoint paths between every pair of vertices.

gatecse-2008 graph-connectivity normal

Answer key

2.5.20 Graph Connectivity: GATE CSE 2013 | Question: 26

The line graph $L(G)$ of a simple graph G is defined as follows:

- There is exactly one vertex $v(e)$ in $L(G)$ for each edge e in G.
- For any two edges e and e' in G, $L(G)$ has an edge between $v(e)$ and $v(e')$, if and only if e and e' are incident with the same vertex in G.

Which of the following statements is/are TRUE?

- (P) The line graph of a cycle is a cycle.
- (Q) The line graph of a clique is a clique.
- (R) The line graph of a planar graph is planar.
- (S) The line graph of a tree is a tree.

A. P only
B. P and R only
C. R only
D. P, Q and S only

gatecse-2013 graph-theory normal graph-connectivity

Answer key

2.5.21 Graph Connectivity: GATE CSE 2014 | Set 1 | Question: 51

Consider an undirected graph G where self-loops are not allowed. The vertex set of G is $\{(i, j) \mid 1 \leq i \leq 12, 1 \leq j \leq 12\}$. There is an edge between (a, b) and (c, d) if $|a-c| \leq 1$ and

$|b-d| \leq 1$. The number of edges in this graph is______.

gatecse-2014-set1 graph-theory numerical-answers normal graph-connectivity

Answer key

2.5.22 Graph Connectivity: GATE CSE 2014 | Set 2 | Question: 3

The maximum number of edges in a bipartite graph on 12 vertices is____

gatecse-2014-set2 graph-theory graph-connectivity numerical-answers normal

Answer key

2.5.23 Graph Connectivity: GATE CSE 2014 | Set 3 | Question: 51

If G is the forest with n vertices and k connected components, how many edges does G have?

A. $\lfloor \frac{n}{k} \rfloor$ B. $\lceil \frac{n}{k} \rceil$ C. $n-k$ D. $n-k+1$

gatecse-2014-set3 graph-theory graph-connectivity normal

Answer key

2.5.24 Graph Connectivity: GATE CSE 2015 | Set 2 | Question: 50

In a connected graph, a bridge is an edge whose removal disconnects the graph. Which one of the following statements is true?

A. A tree has no bridges
B. A bridge cannot be part of a simple cycle
C. Every edge of a clique with size ≥ 3 is a bridge (A clique is any complete subgraph of a graph)
D. A graph with bridges cannot have cycle

gatecse-2015-set2 graph-theory graph-connectivity easy

Answer key

2.5.25 Graph Connectivity: GATE CSE 2019 | Question: 12

Let G be an undirected complete graph on n vertices, where $n > 2$. Then, the number of different Hamiltonian cycles in G is equal to

A. $n!$ B. $(n-1)!$ C. 1 D. $\frac{(n-1)!}{2}$

gatecse-2019 engineering-mathematics discrete-mathematics graph-theory graph-connectivity one-mark

Answer key

2.5.26 Graph Connectivity: GATE CSE 2019 | Question: 38

Let G be any connected, weighted, undirected graph.

I. G has a unique minimum spanning tree, if no two edges of G have the same weight.
II. G has a unique minimum spanning tree, if, for every cut of G, there is a unique minimum-weight edge crossing the cut.

Which of the following statements is/are TRUE?

A. I only B. II only C. Both I and II D. Neither I nor II

gatecse-2019 engineering-mathematics discrete-mathematics graph-theory graph-connectivity two-marks minimum-spanning-tree algorithms graph-algorithms

Answer key

2.5.27 Graph Connectivity: GATE CSE 2021 | Set 1 | Question: 36

Let $G=(V,E)$ be an undirected unweighted connected graph. The *diameter* of G is defined as:

$$\text{diam}(G) = \max_{u,v \in V} \{\text{the length of shortest path between } u \text{ and } v\}$$

Let M be the adjacency matrix of G.

Define graph G_2 on the same set of vertices with adjacency matrix N, where

$$N_{ij} = \begin{cases} 1 & \text{if } M_{ij} > 0 \text{ or } P_{ij} > 0, \text{ where } P = M^2 \\ 0 & \text{otherwise} \end{cases}$$

Which one of the following statements is true?

A. $\text{diam}(G_2) \leq \lceil \text{diam}(G)/2 \rceil$
B. $\lceil \text{diam}(G)/2 \rceil < \text{diam}(G_2) < \text{diam}(G)$
C. $\text{diam}(G_2) = \text{diam}(G)$
D. $\text{diam}(G) < \text{diam}(G_2) \leq 2\,\text{diam}(G)$

gatecse-2021-set1 graph-theory graph-connectivity two-marks

Answer key

2.5.28 Graph Connectivity: GATE CSE 2022 | Question: 20

Consider a simple undirected graph of 10 vertices. If the graph is disconnected, then the maximum number of edges it can have is ______________ .

gatecse-2022 numerical-answers graph-theory graph-connectivity one-mark

Answer key

2.5.29 Graph Connectivity: GATE CSE 2022 | Question: 27

Consider a simple undirected unweighted graph with at least three vertices. If A is the adjacency matrix of the graph, then the number of 3–cycles in the graph is given by the trace of

A. A^3
B. A^3 divided by 2
C. A^3 divided by 3
D. A^3 divided by 6

gatecse-2022 graph-theory graph-connectivity two-marks

Answer key

2.5.30 Graph Connectivity: GATE CSE 2022 | Question: 42

Which of the properties hold for the adjacency matrix A of a simple undirected unweighted graph having n vertices?

A. The diagonal entries of A^2 are the degrees of the vertices of the graph.
B. If the graph is connected, then none of the entries of $A^{n-1} + I_n$ can be zero.
C. If the sum of all the elements of A is at most $2(n-1)$, then the graph must be acyclic.
D. If there is at least a 1 in each of A's rows and columns, then the graph must be connected.

gatecse-2022 graph-theory graph-connectivity multiple-selects two-marks

Answer key

2.5.31 Graph Connectivity: GATE CSE 2024 | Set 1 | Question: 24

The number of spanning trees in a *complete* graph of 4 vertices labelled $\mathrm{A}, \mathrm{B}, \mathrm{C}$, and D is _________.

gatecse-2024-set1 numerical-answers graph-theory graph-connectivity one-mark

Answer key

2.5.32 Graph Connectivity: GATE CSE 2024 | Set 2 | Question: 41

Let G be an undirected connected graph in which every edge has a positive integer weight. Suppose that every spanning tree in G has *even* weight. Which of the following statements is/are TRUE for *every* such graph G?

A. All edges in G have even weight
B. All edges in G have even weight $\mathbf{OR}$ all edges in G have odd weight
C. In each cycle C in G, all edges in C have even weight
D. In each cycle C in G, either all edges in C have even weight OR all edges in C have odd weight

gatecse-2024-set2 graph-theory multiple-selects graph-connectivity two-marks

Answer key☟

2.5.33 Graph Connectivity: GATE CSE 2024 | Set 2 | Question: 7

Let A be the adjacency matrix of a simple undirected graph G. Suppose A is its own inverse. Which one of the following statements is *always* TRUE?

A. G is a cycle
B. G is a perfect matching
C. G is a complete graph
D. There is no such graph G

gatecse-2024-set2 graph-theory graph-connectivity one-mark

Answer key☟

2.5.34 Graph Connectivity: GATE IT 2004 | Question: 37

What is the number of vertices in an undirected connected graph with 27 edges, 6 vertices of degree $2, 3$ vertices of degree 4 and remaining of degree 3?

A. 10
B. 11
C. 18
D. 19

gateit-2004 graph-theory graph-connectivity normal

Answer key☟

2.5.35 Graph Connectivity: GATE IT 2004 | Question: 5

What is the maximum number of edges in an acyclic undirected graph with n vertices?

A. $n-1$
B. n
C. $n+1$
D. $2n-1$

gateit-2004 graph-theory graph-connectivity normal

Answer key☟

2.5.36 Graph Connectivity: GATE IT 2005 | Question: 56

Let G be a directed graph whose vertex set is the set of numbers from 1 to 100. There is an edge from a vertex i to a vertex j iff either $j = i + 1$ or $j = 3i$. The minimum number of edges in a path in G from vertex 1 to vertex 100 is

A. 4
B. 7
C. 23
D. 99

gateit-2005 graph-theory graph-connectivity normal

Answer key☟

2.5.37 Graph Connectivity: GATE IT 2006 | Question: 11

If all the edge weights of an undirected graph are positive, then any subset of edges that connects all the vertices and has minimum total weight is a

A. Hamiltonian cycle
B. grid
C. hypercube
D. tree

gateit-2006 graph-theory graph-connectivity easy

Answer key☟

2.5.38 Graph Connectivity: GATE IT 2007 | Question: 25

What is the largest integer m such that every simple connected graph with n vertices and n edges contains at least m different spanning trees ?

A. 1
B. 2
C. 3
D. n

gateit-2007 graph-theory graph-connectivity tricky

Answer key

2.5.39 Graph Connectivity: GATE IT 2008 | Question: 27

G is a simple undirected graph. Some vertices of G are of odd degree. Add a node v to G and make it adjacent to each odd degree vertex of G. The resultant graph is sure to be

A. regular
B. complete
C. Hamiltonian
D. Euler

gateit-2008 graph-theory graph-connectivity normal

Answer key

2.5.40 Graph Connectivity: GATE IT 2008 | Question: 4

What is the size of the smallest MIS (Maximal Independent Set) of a chain of nine nodes?

A. 5
B. 4
C. 3
D. 2

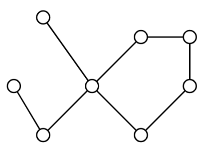
gateit-2008 normal graph-connectivity

Answer key

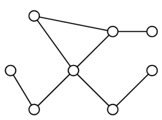
2.6 Graph Isomorphism (4)

2.6.1 Graph Isomorphism: GATE CSE 2012 | Question: 26

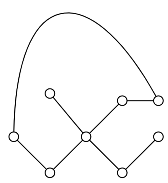
Which of the following graphs is isomorphic to



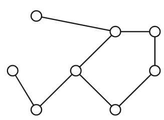
A.



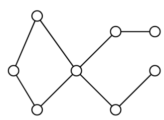
B.



C.



D.



gatecse-2012 graph-theory graph-isomorphism normal

Answer key

2.6.2 Graph Isomorphism: GATE CSE 2014 | Set 2 | Question: 51

A cycle on n vertices is isomorphic to its complement. The value of n is _____.

gatecse-2014-set2 graph-theory numerical-answers normal graph-isomorphism

Answer key

2.6.3 Graph Isomorphism: GATE CSE 2015 | Set 2 | Question: 28

A graph is self-complementary if it is isomorphic to its complement. For all self-complementary graphs on n vertices, n is

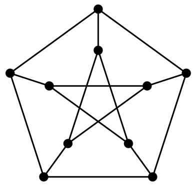
A. A multiple of 4
B. Even
C. Odd
D. Congruent to 0 mod 4, or, 1 mod 4.

gatecse-2015-set2 graph-theory graph-isomorphism

Answer key

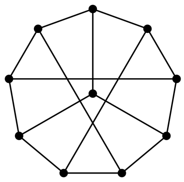
2.6.4 Graph Isomorphism: GATE CSE 2022 | Question: 40

The following simple undirected graph is referred to as the Peterson graph.



Which of the following statements is/are TRUE?

A. The chromatic number of the graph is $3.$
B. The graph has a Hamiltonian path.
C. The following graph is isomorphic to the Peterson graph.



D. The size of the largest independent set of the given graph is $3.$ (A subset of vertices of a graph form an independent set if no two vertices of the subset are adjacent.)

gatecse-2022 graph-theory graph-isomorphism multiple-selects two-marks

Answer key

2.7 Graph Matching (2)

2.7.1 Graph Matching: GATE CSE 2003 | Question: 36

How many perfect matching are there in a complete graph of 6 vertices?

A. 15
B. 24
C. 30
D. 60

gatecse-2003 graph-theory graph-matching normal

Answer key

2.7.2 Graph Matching: GATE CSE 2026 | Set 1 | Question: 47

Let G be an undirected graph, which is a path on 8 vertices. The number of matchings in G is _____. (answer in integer)

gatecse-2026-set1 graph-theory graph-matching numerical-answers two-marks

Answer key

2.8 Graph Planarity (13)

2.8.1 Graph Planarity: GATE CSE 1987 | Question: 2e

State whether the following statement is TRUE or FALSE:

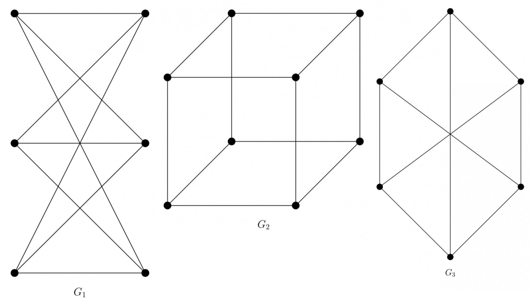
There is a linear-time algorithm for testing the planarity of finite graphs.

gate1987 graph-theory graph-planarity true-false

Answer key

2.8.2 Graph Planarity: GATE CSE 1989 | Question: 3-vi

Which of the following graphs is/are planar?



gate1989 normal graph-theory graph-planarity descriptive

Answer key

2.8.3 Graph Planarity: GATE CSE 1990 | Question: 3-xi

A graph is planar if and only if,

A. It does not contain a subgraph homeomorphic to k_5 and $k_{3,3}$.
B. It does not contain a subgraph isomorphic to k_5 and $k_{3,3}$.
C. It does not contain a subgraph isomorphic to k_5 or $k_{3,3}$
D. It does not contain a subgraph homeomorphic to k_5 or $k_{3,3}$.

gate1990 normal graph-theory graph-planarity multiple-selects

Answer key

2.8.4 Graph Planarity: GATE CSE 1992 | Question: 01,x

Maximum number of edges in a planar graph with n vertices is _____

gate1992 graph-theory graph-planarity easy fill-in-the-blanks

Answer key

2.8.5 Graph Planarity: GATE CSE 1992 | Question: 02,viii

A non-planar graph with minimum number of vertices has

A. 9 edges, 6 vertices
B. 6 edges, 4 vertices
C. 10 edges, 5 vertices
D. 9 edges, 5 vertices

gate1992 graph-theory normal graph-planarity

Answer key

2.8.6 Graph Planarity: GATE CSE 2005 | Question: 10

Let G be a simple connected planar graph with 13 vertices and 19 edges. Then, the number of faces in the planar embedding of the graph is:

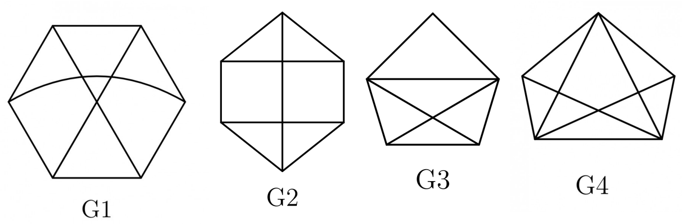
A. 6
B. 8
C. 9
D. 13

gatecse-2005 graph-theory graph-planarity

Answer key

2.8.7 Graph Planarity: GATE CSE 2005 | Question: 47

Which one of the following graphs is NOT planar?



A. G1
B. G2
C. G3
D. G4

gatecse-2005 graph-theory graph-planarity normal

Answer key

2.8.8 Graph Planarity: GATE CSE 2008 | Question: 23

Which of the following statements is true for every planar graph on n vertices?

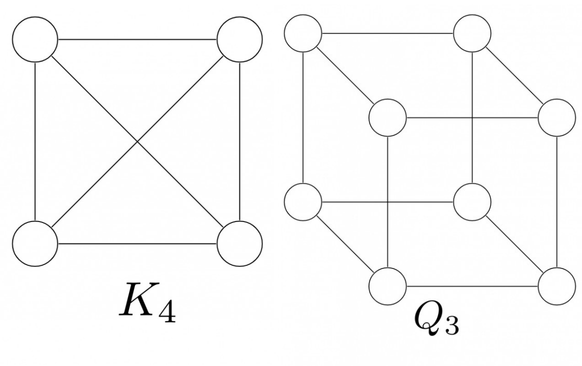
A. The graph is connected
B. The graph is Eulerian
C. The graph has a vertex-cover of size at most $\frac{3n}{4}$
D. The graph has an independent set of size at least $\frac{n}{3}$

gatecse-2008 graph-theory normal graph-planarity

Answer key

2.8.9 Graph Planarity: GATE CSE 2011 | Question: 17

K4 and Q3 are graphs with the following structures.



Which one of the following statements is **TRUE** in relation to these graphs?

A. K4 is a planar while Q3 is not
B. Both K4 and Q3 are planar
C. Q3 is planar while K4 is not
D. Neither K4 nor Q3 is planar

gatecse-2011 graph-theory graph-planarity normal

Answer key

2.8.10 Graph Planarity: GATE CSE 2012 | Question: 17

Let G be a simple undirected planar graph on 10 vertices with 15 edges. If G is a connected graph, then the number of **bounded** faces in any embedding of G on the plane is equal to

A. 3
B. 4
C. 5
D. 6

gatecse-2012 graph-theory graph-planarity normal

Answer key

2.8.11 Graph Planarity: GATE CSE 2014 | Set 3 | Question: 52

Let δ denote the minimum degree of a vertex in a graph. For all planar graphs on n vertices with $\delta \geq 3$, which one of the following is **TRUE**?

A. In any planar embedding, the number of faces is at least $\frac{n}{2}+2$

B. In any planar embedding, the number of faces is less than $\frac{n}{2}+2$
C. There is a planar embedding in which the number of faces is less than $\frac{n}{2}+2$
D. There is a planar embedding in which the number of faces is at most $\frac{n}{\delta+1}$

gatecse-2014-set3 graph-theory graph-planarity normal

Answer key☟

2.8.12 Graph Planarity: GATE CSE 2015 | Set 1 | Question: 54

Let G be a connected planar graph with 10 vertices. If the number of edges on each face is three, then the number of edges in G is_______________.

gatecse-2015-set1 graph-theory graph-connectivity normal graph-planarity numerical-answers

Answer key☟

2.8.13 Graph Planarity: GATE CSE 2021 | Set 1 | Question: 16

In an undirected connected planar graph G, there are eight vertices and five faces. The number of edges in G is _________.

gatecse-2021-set1 graph-theory graph-planarity numerical-answers easy one-mark

Answer key☟

2.9 Jaccard Coefficient (1)

2.9.1 Jaccard Coefficient: GATE CSE 2026 | Set 2 | Question: 26

Consider a complete graph K_n with n vertices ($n > 4$). Note that multiple spanning trees can be constructed over K_n. Each of these spanning trees is represented as a set of edges. The Jaccard coefficient between any two sets is defined as the ratio of the size of the intersection of the two sets to the size of the union of the two sets.

Which one of the following options gives the lowest possible value for the Jaccard coefficient between any two spanning trees of K_n ?

A. $\frac{1}{n}$
B. $\frac{1}{2n-3}$
C. 0
D. $\frac{1}{n-1}$

gatecse-2026-set2 graph-theory jaccard-coefficient two-marks

Answer key☟

Answer Keys

2.1.1	D	2.1.2	D	2.1.3	X	2.2.1	N/A	2.2.2	N/A
2.2.3	N/A	2.2.4	D	2.2.5	C	2.2.6	C	2.2.7	B
2.2.8	C	2.2.9	D	2.2.10	C	2.2.11	C	2.2.12	16
2.2.13	A	2.3.1	B;C	2.4.1	D	2.4.2	C	2.4.3	A
2.4.4	4	2.4.5	3	2.4.6	7	2.4.7	A;B	2.4.8	B;C
2.4.9	2	2.4.10	C	2.4.11	B	2.5.1	N/A	2.5.2	N/A
2.5.3	N/A	2.5.4	N/A	2.5.5	N/A	2.5.6	C;D	2.5.7	C
2.5.8	N/A	2.5.9	N/A	2.5.10	B	2.5.11	D	2.5.12	N/A
2.5.13	B	2.5.14	C	2.5.15	B	2.5.16	A	2.5.17	B
2.5.18	C	2.5.19	D	2.5.20	A	2.5.21	506	2.5.22	36
2.5.23	C	2.5.24	B	2.5.25	C;D	2.5.26	C	2.5.27	A
2.5.28	36	2.5.29	D	2.5.30	A	2.5.31	16	2.5.32	D
2.5.33	B	2.5.34	D	2.5.35	A	2.5.36	B	2.5.37	D

2.5.38	C	2.5.39	D	2.5.40	C	2.6.1	B	2.6.2	5
2.6.3	D	2.6.4	A;B;C	2.7.1	A	2.7.2	34 : 34	2.8.1	TBA
2.8.2	N/A	2.8.3	D	2.8.4	N/A	2.8.5	C	2.8.6	B
2.8.7	A	2.8.8	C	2.8.9	B	2.8.10	D	2.8.11	A
2.8.12	24	2.8.13	11 : 11	2.9.1	C				